

HANDS ON SUBNETTING PRACTICAL

1. Given a network address of 192.168.10.0/24, divide it into 4 equal subnets and write down the subnet mask, network address, and broadcast address for each subnet.

- Network Address: 192.168.10.0/24
- Required Subnets: 4

Step 1: Calculate the New Subnet Mask

- Original subnet: /24
- To create 4 subnets, borrow 2 bits (since $2^2 = 4$).
- New CIDR: /26
- Subnet Mask: 255.255.255.192

Each subnet contains:

- 64 IP addresses
- 62 usable host addresses

Subnet	Network Address	Subnet Mask	First Host	Last Host	Broadcast Address
1	192.168.10.0/26	255.255.255.192	192.168.10.1	192.168.10.62	192.168.10.63
2	192.168.10.64/26	255.255.255.192	192.168.10.65	192.168.10.126	192.168.10.127
3	192.168.10.128/26	255.255.255.192	192.168.10.129	192.168.10.190	192.168.10.191
4	192.168.10.192/26	255.255.255.192	192.168.10.193	192.168.10.254	192.168.10.255

2. For one of the subnets you created, assign valid IP addresses to 5 devices (such as a router, two laptops, a printer, and a mobile) and list which IP goes to which device.

Hint: Make sure not to use the network or broadcast address for any device.



To divide the 192.168.10.0/24 network into 4 equal subnets.

Given

- Network Address: 192.168.10.0/24
- Required Subnets: 4

Step 1: Calculate the Subnet Mask

Number of required subnets = 4

Borrowed bits = 2

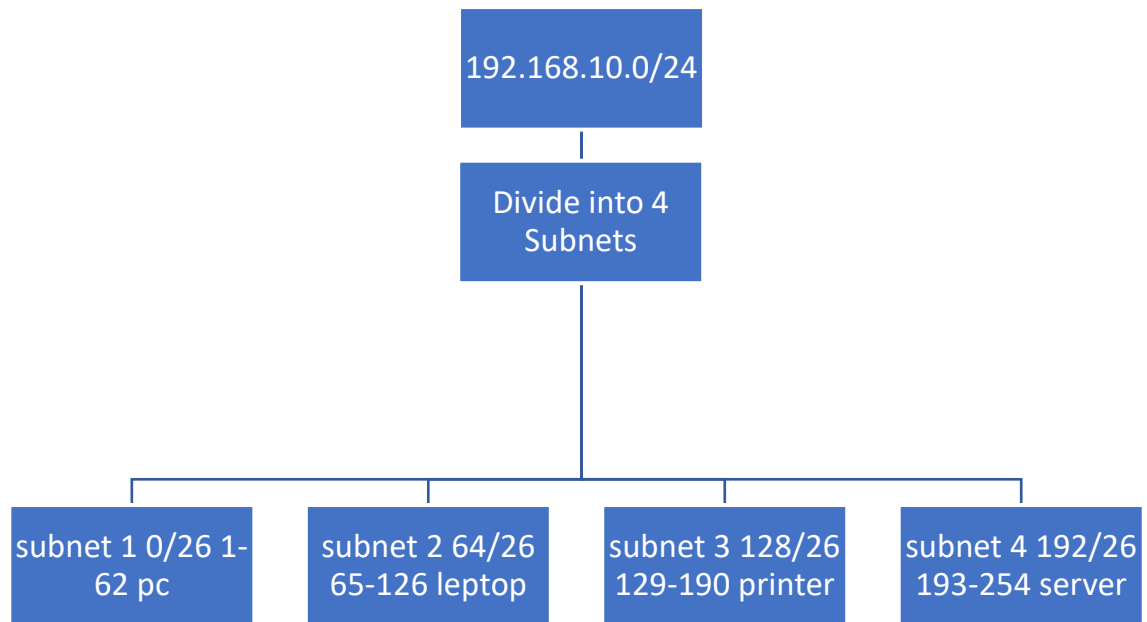
New Prefix = /26

Subnet Mask = 255.255.255.192

Subnet Details

Subnet No.	Subnet Mask	Network Address	Broadcast Address
1	255.255.255.192 (/26)	192.168.10.0	192.168.10.63
2	255.255.255.192 (/26)	192.168.10.64	192.168.10.127
3	255.255.255.192 (/26)	192.168.10.128	192.168.10.191
4	255.255.255.192 (/26)	192.168.10.192	192.168.10.255

3. Draw a simple diagram (hand-drawn or digital) showing your subnet plan: label each subnet, the range of IPs, and where each device is connected.



4. Document step-by-step how you calculated the subnet mask, number of hosts per subnet, and assigned IP addresses for your plan, as if you are explaining it to a friend who is new to subnetting.



Step 1: Start with the Given Network

The given network address is 192.168.10.0/24. The /24 means the first 24 bits are used for the network part, and the remaining bits are available for host addresses.

Step 2: Divide the Network into 4 Equal Subnets

We need 4 equal subnets.

Using the formula:

$$2^2 = 4$$

So, I borrowed 2 bits from the host portion.

New prefix:

$$24 + 2 = /26$$

Therefore, the new subnet mask becomes:

255.255.255.192 (/26)

Step 3: Calculate the Number of Hosts

A /26 subnet leaves 6 host bits.

Total IP addresses in each subnet:

$$2^6 = 64$$

Out of these, the first address is the Network Address and the last address is the Broadcast Address.

So,

$$\text{Usable Host Addresses} = 64 - 2 = 62$$

Each subnet can support 62 devices.

Step 4: Create the Four Subnets

Since the block size is 64, the subnet addresses increase by 64 each time.

- Subnet 1 → 192.168.10.0
- Subnet 2 → 192.168.10.64
- Subnet 3 → 192.168.10.128
- Subnet 4 → 192.168.10.192

Step 5: Assign IP Addresses

I assigned one sample device in each subnet.

Device	Assigned IP Address
PC	192.168.10.10
Laptop	192.168.10.70
Printer	192.168.10.140
Server	192.168.10.200

Each IP address belongs to its respective subnet and does not use the network or broadcast address.